



PegShield

github.com/fexx301/pegshield

Parametric USDC depeg protection, settled by Attestcoin.

A user buys fixed-coverage protection on Creditcoin CC3. When the pinned Ethereum USDC/USD aggregator publishes two qualifying below-threshold updates, anyone submits the two Attestcoin proofs in **one atomic CC3 transaction** — and the pool pays the exact locked coverage to the beneficiary stored at purchase.

The adjuster becomes an on-chain fact.

BUIDL CTC 2026 Fall

Attestcoin Protocol

Cross-chain DeFi

Live on testnet

THE PROBLEM

A payout decided by an operator has only moved the adjuster problem.

Traditional insurance waits on a human adjuster. On-chain versions replace the adjuster with an oracle operator — who can be **wrong, gamed, or down at the exact moment a claim must settle**. Whoever is waiting for the payout only has the operator's word.

Adjuster

Slow, contestable, opaque. The beneficiary holds a promise, not a fact.

Oracle operator

Centralized assertion. The same trust, relocated — with a nicer API.

Bridge-based cover

Locks assets in vaults: the target of billions in hacks.

THE ANSWER

An oracle can assert a depeg. PegShield **verifies the receipt.**

The payout is decided by the Ethereum receipt of the Chainlink transmission itself — inclusion-proven and continuity-anchored by Attestcoin. No settlement operator can assert, alter, or withhold what the reference feed transmitted.

Kept

- ♦ The industry-grade reference feed (pinned USDC/USD aggregator)
- ♦ Fixed coverage and locked beneficiary, chosen at purchase

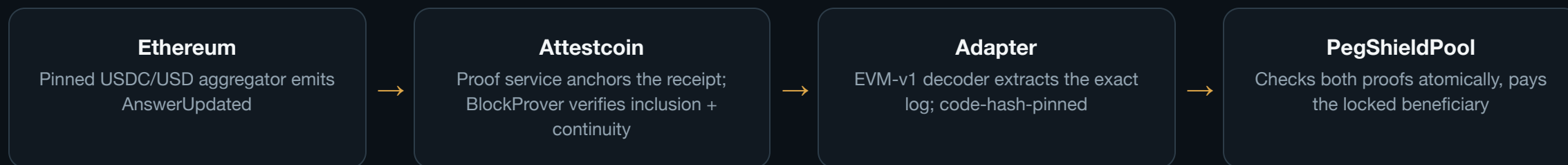
Removed

- ♦ Assertion trust: no one tells the contract what the feed said
- ♦ Relay trust: submitter cannot influence a single field of the decision

The CC3 contract re-derives the answer, round, and timestamp from the authenticated log. The relayer submits opaque proof bytes and nothing else.

HOW IT WORKS

One transaction, every check.



Both proofs settle in a single `submitClaim` — a permissionless relayer cannot pin an unusably late first observation, because a failed or maliciously ordered pair leaves the policy untouched.

- ✓ Receipt succeeded on source chain
- ✓ Positive answer, strictly below trigger
- ✓ Confirmation strictly later (round + time)
- ✓ Per-policy replay protection on attested identity
- ✓ Correct chain key + emitter + event topic
- ✓ Both observations inside the coverage window
- ✓ Minimum breach interval met
- ✓ Effects before transfer; exact token-delta accounting

Not a single event read.

A funded, atomic, two-proof settlement.

ATTESTCOIN COMPONENT	WHERE	WHAT IT SETTLES
BlockProver precompile 0x...FD2	Adapter's only window — <code>verifySourceLog</code>	Transaction inclusion (Merkle) + chain continuity, per proof
EVM-v1 decoder 0x731c... (code-hash pinned)	Same adapter	Receipt status, emitter, exact log at a bounds-checked position
ChainInfo precompile 0x...FD3	Deployment preflight	Dependency set is live and attesting before any broadcast
Official proof service (batch + single)	Worker CLI + web claim service	Fresh continuity anchors; per-tx single-item paths for distant blocks
Discovery lock	Every consumer	Chain IDs, official addresses, code hashes, event topic — pinned, not caller-configurable

Same-day purchase → payout: 14 minutes.

08:01

Policy purchased (100 tUSD coverage, 2.5 tUSD premium)

1 tx

Atomic claim: both Attestcoin proofs verified and paid in one transaction

100

tUSD paid to the beneficiary locked at purchase — exact to the wei

FACT	VALUE / RECEIPT
Source rounds (Ethereum)	1187 (0.99986193, 07:47:15 UTC) + 1188 (0.99986417, 08:40:35 UTC) — exactly 3,600 s apart
Atomic claim + payout	creditcoin-testnet.blockscout.com/tx/0x575030ff...c2120c8 · 616,966 gas · 4 events
Accounting	reserved 1,000,000,000 → 0 · accounted 1,002,500,000 → 902,500,000
First end-to-end settlement (v3, 2026-09-04)	Rounds 1171/1172 · 835,156 gas · same exact-payout shape

Evidence-first, all the way down.

58 Foundry tests

- ♦ 256-run invariant fuzzing (reserved $\leq$ accounted $\leq$ balance)
- ♦ ERC-20 callback/reentrancy regression
- ♦ Named grieving-vector regression: late-first-observation pinning

29 worker + 28 web tests

- ♦ Artifact integrity, strict-below trigger boundary, rate limits, 404 mapping
- ♦ Playwright e2e: fixture walkthrough, prepare-failure recovery

Gates that bite

- ♦ CI green (6 consecutive): contracts, typescript, schemas, e2e
- ♦ ABI drift checks parse real TS exports, compare full signatures
- ♦ Schema-validated manifests; pinned dependency code hashes

A 28-document evidence chain (P01–P29) records every discovery, deployment, incident, and settlement with receipts — including the failures.

What we learned about the **Attestcoin Protocol**.

Proof lifetime

A committed proof survived BlockProver verification **61 hours** after generation. A single-proof continuity path later failed as the attestation window advanced; a fresh batch request restored the same round.

Shared-anchor limits

A two-transaction proof batch shares one continuity anchor — valid for nearby headers only. For distant blocks, the later proof fails with **Merkle root mismatch**; per-transaction single-item batches verify. Confirmed at the BlockProver level.

CC3 estimateGas

eth_estimateGas is intermittently unstable (observed ~50% of actual need — a broadcast ran out of gas at 110,352 vs a fresh estimate of 226,245). Broadcast with explicit gas limits.

Operational rules any Attestcoin builder needs — documented with receipts, published back to the ecosystem.

The boundary, stated precisely.

Removed

- ♦ Settlement-layer operators: no one between the Ethereum receipt and the CC3 payout can assert, alter, or withhold
- ♦ Relayer, browser, and server trust
- ♦ Caller-supplied price, round, timestamp, beneficiary, feed, or hash

Retained — by design

- ♦ Reference-feed quality: a wrong aggregator answer still pays, exactly as real parametric products insure the index, not ground truth
- ♦ Aggregation between the two observations is not proven uninterrupted

Feed choice and window are product parameters pinned per policy at purchase — documented, not hidden. That is the difference between a trust boundary and a trust claim.

Cover the collateral, unlock the credit.

A CC3 lending market names itself as the policy beneficiary at purchase. Depeg cover on the USDC collateral is what lets undercollateralized lending against it survive the tail event — protection settles natively where the credit application needs liquidity.

Product

- ♦ Fixed coverage, premium in bps (ceil-rounded on-chain)
- ♦ Underwriter-funded pool, exact reserve accounting
- ♦ Immutable per-policy terms; expiry; one-way product disable

Stack

- ♦ Contracts: pool + adapter (Foundry, 0.8.28)
- ♦ Worker: discovery, proof build, claim calldata CLI
- ♦ Web: wallet-gated dashboard + public discovery API

Verify everything

- ♦ github.com/fexx301/pegshield
- ♦ README → "Why Attestcoin" + verify-everything table
- ♦ docs/evidence/P29 — same-day settlement with receipts

Buy fixed coverage. Prove the depeg twice. Let Creditcoin pay.